Algo ri

26/8/22

Advantages:

i) Determining the total cost, variable cost & fixed cost at a given level of alternative.

ii) Finding out BE output & sales

iii) Understanding the cost, volume & profit relationship

iv) Making inter-firm comparison.

v) Enforcing cost control. ✓

Limitations:

i) difficult to segregate the cost into fixed & variable component as both of them varies to a certain extend after production reaches to a certain level.

ii) The application of BE analysis to a multiproduct firm is very difficult. A lot of complicated complications are involved.

iii) BEP has only limited importance. At best, it would indulge management in cost reduction over times of dull business period.

iv) BEP provides neither a std. performance nor a guide for executive decision.

* **Estimation Techniques:**

1. Algorithmic cost modeling
2. Expert judgement
3. Estimation by analogy
4. Parkinson's Law ✓

~~5. pricing to win~~

(5) pricing to win

Algorithmic Cost modeling:

↳ based on historical cost information that relates some s/w metric to the project cost is used.

Parkinson's Law:

- Work expands to fill the time available.
- Available resources determine the cost rather than by objective assessment.
- If the s/w has to be delivered in 12 months & 5 people are available, the effort required is estimated to be 60 persons month.

* COCOMO Model:

- Dr. Barry Boehm's.
- It is an algorithmic model developed based on project experience.
- It is heuristic in nature.

Advantages:

i) Objective, repeatable
ii) has modifiable & analyzable formulas.
iii) Efficient & able to support sensitivity analysis
iv) objectively calibrated to previous experience

Disadv.

i) Unable to deal with exceptional conditions.
ii) Poor sizing i/ps & inaccurate cost driver rating will result in inaccurate estimation
iii) Some experience & factors can not be easily quantified.

Assumptions of COCOMO

i) System development should use waterfall process.
ii) All s/w development start from scratch.

iii) Development time doesn't increase linearly with product size for larger products.

iv) Development time is roughly the same for all the three categories of products.

$$\boxed{Effort = a_1 * (KLOC)^{A} \; PM}$$

$$\boxed{T_{dev} = b_1 * (Effort)^{B} \; months}$$

a) Simple/organic Model with project group:
- Relatively small group
- working to develop well understood applications.

b) Moderate/semidetached model
- Consists of mixture of experienced & inexperienced staff.

c) Embedded model emphasizes on the type of erogonomics:
- The s/w is strongly coupled to complex hardware or real time system.

Project	a_1	A	b_1	B
Simple/organic	2.4	1.05	2.5	~~0.38~~
Moderate	3.0	1.12	2.5	0.35
Embedded	3.6	~~1~~. 2	2.5	0.32

Development Size Effort:

Organic: $Effort = 2.4(KLOC)^{1.05}$ PM

Moderate: $Effort = 3.0(KLOC)^{1.12}$ PM

Embedded: $Effort = 3.6(KLOC)^{1.20}$ PM

Estimation of development time:

Organic : $T_{dev} = 2.5\,(Effort)^{0.38}$ months

Moderate : $T_{dev} = 2.5\,(Effort)^{0.35}$ months

Embedded : $T_{dev} = 2.5\,(Effort)^{0.32}$ months

$$\text{Productivity}\ (P) = \left[\frac{KLOC}{Effort}\right]\ \frac{KLOC}{PM}$$

1 Eg: LOC = 62000, avg. salary of s/w engineer = 20K per man month.

a) effort = ?

b) T_{dev} = ?

c) cost required = ?

} organic project